

A Note on Domination and Total Domination in Prisms

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Abstract

Recently, Azarija et al. considered the prism $G \square K_2$ of a graph G and showed that $\gamma_t(G \square K_2) = 2\gamma(G)$ if G is bipartite, where $\gamma_t(G)$ and $\gamma(G)$ are the total domination number and the domination number of G . In this note, we give a simple proof and observe that there are similar results for other pairs of parameters. We also answer a question from that paper and show that for all graphs $\gamma_t(G \square K_2) \geq \frac{4}{3}\gamma(G)$, and this bound is tight.

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1 Introduction

In this note, we study parameters in prisms of graphs. For graph G , the *prism* of G is defined by taking two disjoint copies G_1 and G_2 of G , called layers, and adding an edge between each pair of corresponding vertices. For each vertex v in G we denote its equivalent in G_i by v_i , and refer to the vertices v_1 and v_2 as partners. We will denote the prism by $G \square K_2$.

The relationship between domination parameters in the graph and its prism have been studied in [2, 6, 9, 10, 11, 12] inter alia. It is immediate that the projection of a dominating set of the prism onto either layer dominates G . Similarly, taking the same dominating set in each layer provides a total dominating set of the prism. Thus, denoting the domination number and the total domination number of G by $\gamma(G)$ and $\gamma_t(G)$, respectively, we have the following.

Observation 1.1 *If G is a graph and H its prism, then $\gamma(G) \leq \gamma_t(H) \leq 2\gamma(G)$.*

Extending the result for trees given in [9], recently Azarija et al. [1] showed that if G is a bipartite graph, then $\gamma_t(G \square K_2) = 2\gamma(G)$. We provide here a simple proof and observe that this idea extends to some other pairs of related parameters.

Further, they showed that the bipartite condition is essential, by constructing an infinite family of connected graphs G satisfying $\gamma_t(G \square K_2) = \frac{3}{2}\gamma(G)$. Answering a question from that paper, we show that there are graphs with $\gamma_t(G \square K_2) = \frac{4}{3}\gamma(G)$ and the value $4/3$ is best possible.

2 Open and Closed Parameters

Both domination and total domination have been generalized. Recall that $N(v)$ denotes the open neighborhood and $N[v]$ the closed neighborhood of vertex v . A *k-tuple dominating set* S requires that $|S \cap N[v]| \geq k$ for all vertices v (see [5]); and a *k-tuple total dominating set* that $|S \cap N(v)| \geq k$ for all vertices v (see [8]). A *paired dominating set* is a dominating set S such that the subgraph induced by S has a perfect matching (see [7]). The minimum cardinalities of such a set are denoted $\gamma_{\times k}(G)$, $\gamma_{\times k,t}(G)$ and $\gamma_{\text{pr}}(G)$. (Note that $\gamma_{\times 1}(G) = \gamma(G)$ and $\gamma_{\times 1,t}(G) = \gamma_t(G)$, and that $\gamma_t(G) \leq \gamma_{\text{pr}}(G)$.)

There are also parameters dual to domination and total domination. An *open packing* S requires that $|S \cap N(v)| \leq 1$ for all vertices v (see [4]); and a *packing* S requires that $|S \cap N[v]| \leq 1$ for all vertices v . We prove:

Theorem 2.1 *If G is a bipartite graph and H its prism, then*

- (a) $\gamma_{\text{pr}}(H) = 2\gamma(G)$,
- (b) $\gamma_{\times k, t}(H) = 2\gamma_{\times k}(G)$ for all $k \geq 1$, and
- (c) $\rho^o(H) = 2\rho(G)$.

Proof. (a) Let π be the natural projection from the prism H to graph G . Then open neighborhoods project onto closed neighborhoods. That is, for any vertex v in H it holds that $N_G[\pi(v)] = \pi(N_H(v))$. Now, assume that G is bipartite. This means that H is bipartite too, say with bipartition (X, Y) . Then for any vertex x in X , since all its neighbors are in Y , it holds that

$$N_G[\pi(x)] = \pi(N_H(x) \cap Y).$$

In particular, if T is a total dominating set of H , then $N_H(x) \cap Y$ contains a vertex of $T \cap Y$ and so $N_G[\pi(x)]$ contains a vertex of $\pi(T \cap Y)$. Since $\pi(X)$ is all of $V(G)$, it follows that $\pi(Y \cap T)$ is a dominating set of G . Similarly, $\pi(X \cap T)$ is a dominating set of G . Since $Y \cap T$ and $X \cap T$ are disjoint, it follows that $\gamma_t(H) \geq 2\gamma(G)$. On the other hand, the construction uses identical dominating sets of each layer, so $\gamma_{\text{pr}}(H) \leq 2\gamma(G)$. Thus, $2\gamma(G) \leq \gamma_t(H) \leq \gamma_{\text{pr}}(H) \leq 2\gamma(G)$, implying equality throughout this inequality chain.

(b,c) These results follow similarly. If T is a k -tuple total dominating set [open packing], then for each color, the projection of the vertices of T of that color forms a k -tuple dominating set [packing] of G . On the other hand, taking a copy of a k -tuple dominating set [packing] in each layer provides a k -tuple total dominating set [open packing] of the prism. $\square$

One can also establish the above theorem by noting that when G is bipartite, the open neighborhood hypergraph of the prism $G \square K_2$ is isomorphic to two disjoint copies of the closed neighborhood hypergraph of G .

3 Lower Bound on Total Domination

Azarija et al. [1] asked what the smallest the total domination number of the prism could be as a function of the domination number of the original graph. We resolve this.

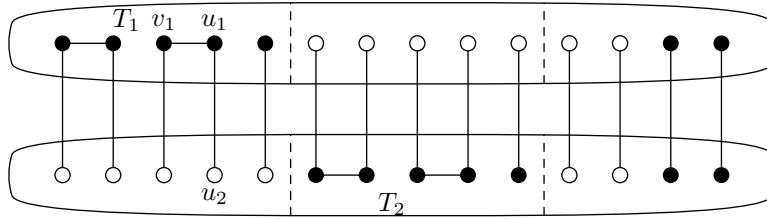
Theorem 3.1 *If G is a graph, then $\gamma_t(G \square K_2) \geq \frac{4}{3}\gamma(G)$, and this bound is tight.*

Proof. We first prove the bound. Let T be a total dominating set of $G \square K_2$. Let T_1 be those vertices in T in G_1 whose partner in G_2 is not in T . Let T'_1 be the subset

of T_1 that have a neighbor in T_1 . Similarly, let T_2 be those vertices in T in G_2 whose partner in G_1 is not in T , and T'_2 be the subset of T_2 that have a neighbor in T_2 .

We will add vertices to T to form a set U such that every vertex is dominated from within their layer; that is, U restricted to $V(G_1)$ dominates G_1 , and U restricted to $V(G_2)$ dominates G_2 .

Consider a vertex u_2 not dominated by T from within G_2 . It follows that its partner u_1 is in T , in particular in T_1 . Further, u_1 has a neighbor v_1 in T . Since u_2 is not dominated from within G_2 , this means that v_2 is not in T , so that v_1 is in T_1 , and in fact, both u_1 and v_1 are in T'_1 .

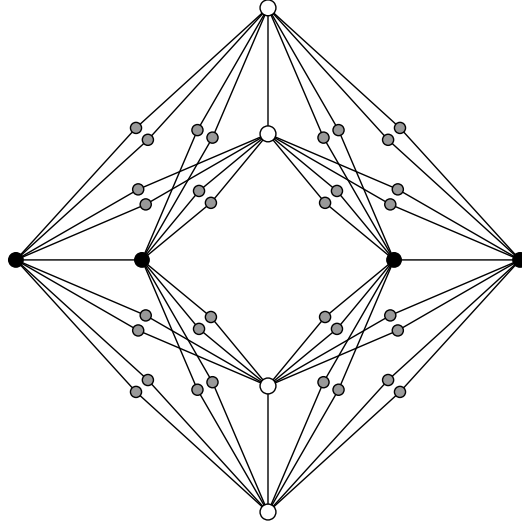


Let D_1 be a minimum dominating set of the subgraph $G_1[T'_1]$ of G_1 induced by the set T'_1 . Note that T'_1 has no isolates. Thus, by Ore's Theorem [13], it follows that $|D_1|$ is at most $|T'_1|/2$. Let D_2 be the set of all vertices w_2 in G_2 whose partner w_1 is in D_1 . Then, since D_1 dominates u_1 , it follows that D_2 dominates u_2 . Or more generally, D_2 dominates every vertex in G_2 not dominated from within G_2 by T .

Similarly, define E_2 as a minimum dominating set of the subgraph $G_2[T'_2]$ of G_2 induced by the set T'_2 , and let E_1 be the set of all vertices w in G_1 whose partner w' is in E_2 . Analogous arguments show that $|E_2|$ is at most $|T'_2|/2$ and that E_1 dominates every vertex in G_1 not dominated from within G_1 by T .

We now form the set U by adding to T all of D_2 and E_1 . By construction, $|U| = |T| + |D_2| + |E_1| \leq |T| + |T'_1|/2 + |T'_2|/2 \leq |T| + |T|/2$. For $i \in \{1, 2\}$, let $U_i = U \cap V(G_i)$ and note that U_i dominates G_i . This implies that $\gamma(G) \leq |U|$, and so $2\gamma(G) \leq |U_1| + |U_2| = |T| + |D_2| + |E_1| \leq 3|T|/2$, or, equivalently, $\gamma(G) \leq \frac{3}{4}|T| = \frac{3}{4}\gamma_t(G \square K_2)$. This establishes the desired lower bound on $\gamma_t(G \square K_2)$.

We show next that the bound is tight. Consider the following construction. For F a graph without isolates define $G(F)$ as follows. Let F' be a disjoint copy of F . Take the two copies and for every pair of vertices $f \in V(F)$ and $f' \in V(F')$, add a set $A_{f,f'}$ of two (or more) independent vertices, whose only neighbors are f and f' . The case $F = 2K_2$ is drawn below, where the white vertices belong to F , the black vertices to F' , and the shaded vertices belong to the added sets $A_{f,f'}$.



Claim 3.2 $\gamma(G(F)) = |V(F)| + \gamma(F)$.

Proof. Let D be a minimum dominating set of $G(F)$. Let $f \in V(F)$ and $f' \in V(F')$, and consider the set $A_{f,f'}$. If neither f nor f' belongs to D , then in order to dominate the vertices in $A_{f,f'}$, we need both the vertices in D . But then we can replace them with f and f' and still have a dominating set. It follows that we may assume that the set D contains at least one of $\{f, f'\}$ for every pair (f, f') .

This means that D contains all of F or all of F' . Say the former. Then, none of the vertices of F' is dominated by a vertex in F . Further, if some vertex of $A_{f,f'}$ is in D , we can replace it by f' and still have a dominating set. That is, we may assume that D restricted to F' dominates F' , and so $|D \cap V(F')| \geq \gamma(F')$. Hence, $|D| = |V(F)| + |D \cap V(F')| \geq |V(F)| + \gamma(F)$. Thus, $\gamma(G(F)) \geq |V(F)| + \gamma(F)$.

But clearly this is achievable, since taking all of F and a minimum dominating set of F' is a dominating set of the graph $G(F)$. ($\square$)

Claim 3.3 $\gamma_t(G(F) \square K_2) \leq 2|V(F)|$.

Proof. Consider the set T consisting of the vertices of F in the first layer and the vertices of F' in the second layer. Then every vertex of F in both layers is dominated (as F had no isolates); similarly with F' . Further, every vertex of every $A_{f,f'}$ is dominated, since we take f in one layer and f' in the other. ($\square$)

We now return to our proof of Theorem 3.1. Let F be a graph of order $2m$ whose domination number is m . For example, $F = mK_2$. Then, by Claim 3.2, $\gamma(G(F)) = 3m$.

By Claim 3.3, $\gamma_t(G(F) \square K_2) \leq 4m$. As shown earlier, for every graph G , we have $\gamma_t(G \square K_2) \geq \frac{4}{3}\gamma(G)$. Consequently,

$$\gamma_t(G(F) \square K_2) = 4m = \frac{4}{3}\gamma(G(F)).$$

This completes the proof of Theorem 3.1. $\square$

4 Lower Bound on Total k -tuple Domination

Theorem 3.1 generalizes.

Theorem 4.1 (a) For graph G and $k \geq 1$, it holds that $\gamma_{\times k, t}(G \square K_2) \geq \frac{4}{3}\gamma_{\times k}(G)$.
(b) This bound is tight for $k \in \{1, 2\}$.
(c) For each $k \geq 3$, this bound is not achievable but is asymptotically tight.

Proof. (a) The proof is a direct generalization of the proof of Theorem 3.1. Let T be a k -tuple total dominating set of the prism. Define sets T_1 and T'_1 as before. Consider a vertex u_2 that is not dominated k times from within its layer. Since u_2 has k neighbors in T , this means that its partner u_1 is in T but u_2 is not. The vertex u_1 has k neighbors in T within its layer, but u_2 does not. Thus u_1 has some neighbor v_1 (say) such that v_1 is in T but its partner v_2 is not. That is, u_1 and v_1 are in T'_1 .

Thereafter, define D_2 as before as the partners of a minimum dominating set of $G_1[T'_1]$. Similarly, define E_1 analogously and let $U = T \cup D_2 \cup E_1$ as before. Note that vertex u_2 is dominated $k - 1$ times within its layer by T . Thus U dominates u_2 k times from within its layer. That is, U can be partitioned into a k -tuple dominating set of each layer, while $|U| \leq \frac{3}{2}|T|$. The bound follows as before.

(b) Construct graph $G_2(C_4)$ as follows. Take two disjoint copies F and F' of the 4-cycle. For every pair f of vertices in F and every pair f' of vertices in F' , create a set $A_{f, f'}$ of (at least) 6 “free” vertices each of whose neighborhood is $f \cup f'$. Add a perfect matching between $V(F)$ and $V(F')$.

Claim 3.2 generalizes: $\gamma_{\times 2, t}(G_2(C_4) \square K_2) \leq 8$.

For the 2-tuple total dominating set T of the prism, take all of F in one layer and all of F' in the other layer. Every free vertex has 2 neighbors in this set. Every vertex of $F \cup F'$ in each layer also has 2 neighbors in T : either both within its layer (the 4-cycle is 2-regular) or one in each layer.

Further: $\gamma_{\times 2}(G_2(C_4)) \geq 6$.

Let D be a 2-tuple dominating set (also known as a double dominating set) of the graph $G_2(C_4)$. If some free vertex, say $w \in A_{f,f'}$, does not have 2 neighbors in D , then D must contain all of $A_{f,f'}$. Therefore, one can redistribute 4 of these to $f \cup f'$ and still have a 2-tuple dominating set. Thus, we may assume that every free vertex has 2 neighbors in D . This means that either D contains all of one of F or F' , say F ; or D is missing at most 2 vertices of $F \cup F'$.

The second case implies that $|D| \geq 2|V(F)| - 2 = 6$. On the other hand, if D contains all of F , then those vertices dominate each vertex in F' only once. So we need to dominate F' once more. This requires at least 2 more vertices, since neither a free vertex nor a vertex of F' dominates all of F' .

(c) Equality in part (a) requires that $|D_2| + |E_1| = |T|/2$. In particular this means that (i) $G_1[T'_1]$ has domination number half its order; and (ii) that $|T'_1| + |T'_2| = |T|$. By the result of Fink et al. [3], Condition (i) implies that $G_1[T'_1]$ has minimum degree at most 2. Condition (ii) implies that $T'_1 \cup T'_2 = T$. Now, a vertex of T'_1 has k neighbors in T , and since there is no edge between T_1 and T_2 , these neighbors must be in T'_1 . That is, $G_1[T'_1]$ has minimum degree at least k , a contradiction.

The bound is asymptotically best possible. For example, let graph R be the join of the complete graph K_{k-1} with r disjoint copies of the graph $G(K_2)$ (an optimal graph for the $k = 1$ case). Then, the k -tuple total domination number of the prism of R is at most $4r + 2(k - 1)$ (by taking all $k - 1$ vertices of the complete graphs in both layers, and a total dominating set from each copy of $G(K_2) \square K_2$); further the k -tuple domination number of R is at least $3r$. $\square$

5 Upper Bound on Open Packing

The question for packing and open packing is much less exciting.

Theorem 5.1 $\rho^o(G \square K_2) \leq 4\rho(G)$, and this bound is tight.

Proof. By general results, it follows that $\rho^o(G \square K_2) \leq 2\rho^o(G) \leq 4\rho(G)$. To show that the bound is tight, consider the Petersen graph P . The Petersen graph has $\rho(P) = 1$ (it has diameter 2) while $\rho^o(P \square K_2) = 4$ (the Petersen graph contains $2K_2$ as an induced subgraph). $\square$

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